$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & 0 & 0 \\
\mathcal{K}^{\#1}_{0^-}{}^{\dagger\alpha\beta\chi} & 0 & 3\,k^2\,\overset{(4)}{\kappa}_{15}+3\,\overset{(2)}{\kappa}_{1}
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}{}^{\dagger\alpha\beta} & \frac{1}{2}\left(-\mathrm{Y}_1\,k^2+4\,\overset{(2)}{\kappa}_{1}\right) & \frac{\mathrm{Y}_2\,k^2+4\,\overset{(2)}{\kappa}_{1}}{2\,\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}{}^{\dagger\alpha\beta} & \frac{\mathrm{Y}_2\,k^2+4\,\overset{(2)}{\kappa}_{1}}{2\,\sqrt{2}} & \frac{1}{2}\left(\mathrm{Y}_3\,k^2+2\,\overset{(2)}{\kappa}_{1}\right) & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\alpha} & 0 & 0 & 0 & 0 \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\alpha} & 0 & 0 & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\alpha\beta} & \frac{3\,k^2\,\overset{(4)}{\kappa}_{1}}{2} & 0 \\
\mathcal{K}^{\#1}_{2^-}{}^{\dagger\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & 0 & 0 \\
\mathcal{J}^{\#1}_{0^-}{}^{\dagger\alpha\beta\chi} & 0 & \frac{1}{3\,(k^2\,\overset{(4)}{\kappa}_{15}+\overset{(2)}{\kappa}_{1})}
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}{}^{\dagger\alpha\beta} & \frac{\mathrm{Y}_3\,k^2+2\,\overset{(2)}{\kappa}_{1}}{2\,\mathrm{Det}\,(1^+)} & \frac{-\mathrm{Y}_2\,k^2-4\,\overset{(2)}{\kappa}_{1}}{2\,\sqrt{2}\,\mathrm{Det}\,(1^+)} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}{}^{\dagger\alpha\beta} & \frac{-\mathrm{Y}_2\,k^2-4\,\overset{(2)}{\kappa}_{1}}{2\,\sqrt{2}\,\mathrm{Det}\,(1^+)} & \frac{-\mathrm{Y}_1\,k^2+4\,\overset{(2)}{\kappa}_{1}}{2\,\mathrm{Det}\,(1^+)} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\alpha} & 0 & 0 & 0 & 0 \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\alpha} & 0 & 0 & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\alpha\beta} & \frac{2}{3\,k^2\,\overset{(4)}{\kappa}_{1}} & 0 \\
\mathcal{J}^{\#1}_{2^-}{}^{\dagger\alpha\beta\chi} & 0 & 0
\end{array}
$$

Abbreviations used in matrices

$$\mathrm{Y}_1 == 3\,\overset{(4)}{\kappa}_{1}-4\,\overset{(4)}{\kappa}_{15}+2\,\overset{(4)}{\kappa}_{3}\;\&\&\;\mathrm{Y}_2 == 4\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_{4}\;\&\&\;\mathrm{Y}_3 == 2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_{3}+\overset{(4)}{\kappa}_{4}\;\&\&$$

$$\mathrm{Det}\,(1^+) == -\frac{1}{8}\,k^2\,(k^2\,(6\,\overset{(4)}{\kappa}_{1}\,(2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_{3}+\overset{(4)}{\kappa}_{4})+(2\,\overset{(4)}{\kappa}_{3}+\overset{(4)}{\kappa}_{4})^2)+12\,\overset{(4)}{\kappa}_{1}\,\overset{(2)}{\kappa}_{1})$$

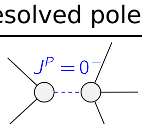
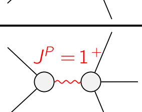
Lagrangian

$$\overset{(2)}{\kappa}_{1}\,\mathcal{K}_{\alpha\beta\chi}\,\mathcal{K}^{\alpha\beta\chi}-2\,\overset{(2)}{\kappa}_{1}\,\mathcal{K}^{\alpha\beta\chi}\,\mathcal{K}_{\beta\alpha\chi}+\overset{(4)}{\kappa}_{1}\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}+$$

$$\overset{(4)}{\kappa}_{3}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta}+\overset{(4)}{\kappa}_{4}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta}-3\,\overset{(4)}{\kappa}_{1}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta}-\overset{(4)}{\kappa}_{3}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta}+$$

$$\frac{1}{2}\,\overset{(4)}{\kappa}_{3}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}+\frac{1}{2}\,\overset{(4)}{\kappa}_{4}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}+\overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}-2\,\overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\,\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1}_{0^+} == 0$	1	$\partial_\beta\mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$
$\mathcal{J}^{\#2\,\alpha}_{1^-} == 0$	3	$\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi} == 0$
$\mathcal{J}^{\#1\,\alpha}_{1^-} == 0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi}+\partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == \partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{2^-} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon}+3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta}+$ $4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} ==$ $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon}+3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi}+2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta}+$ $2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi}+4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon}+3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon}+3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
Total # constraint(s):	12	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$-\frac{\overset{(2)}{\kappa}_{1}}{\overset{(4)}{\kappa}_{15}}$	$-\frac{1}{3\,\overset{(4)}{\kappa}_{15}}$
	3	$-\frac{12\,\overset{(4)}{\kappa}_{1}\,\overset{(2)}{\kappa}_{1}}{6\,\overset{(4)}{\kappa}_{1}\,(2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_{3}+\overset{(4)}{\kappa}_{4})+(2\,\overset{(4)}{\kappa}_{3}+\overset{(4)}{\kappa}_{4})^2}$	$\frac{2\,(6\,\overset{(4)}{\kappa}_{1}{}^2+4\,\overset{(4)}{\kappa}_{1}\,(2\,\overset{(4)}{\kappa}_{3}+\overset{(4)}{\kappa}_{4})+(2\,\overset{(4)}{\kappa}_{3}+\overset{(4)}{\kappa}_{4})^2)}{\overset{(4)}{\kappa}_{1}\,(6\,\overset{(4)}{\kappa}_{1}\,(2\,\overset{(4)}{\kappa}_{15}+\overset{(4)}{\kappa}_{3}+\overset{(4)}{\kappa}_{4})+(2\,\overset{(4)}{\kappa}_{3}+\overset{(4)}{\kappa}_{4})^2)}$
Resolved unitarity condition(s):	(Demonstrably impossible)		